Sriteja Valaboju

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Education

Chaitanya Institute of Technology and Science, Hanamkonda — **August 2018 – June 2022**
Bachelor of Technology in Computer Science Engineering (CGPA: 8.07) — *Warangal, India*

Experience

Airbus Group India Pvt Ltd. — **April 2025 – Present**
Software Engineer — *Bengaluru, India*

- Architected and delivered Augmented BMS, a GenAI-powered conversational chatbot over a corpus of nearly 4,000 interconnected business management documents, serving 80,000 users with 5,000+ monthly active users across Airbus; applied ReAct (Reason + Act) patterns for multi-step query resolution.
- Engineered multi-agent workflows using Google ADK for Advanced Search (Deep Mode) and ABR link-analysis features, incorporating human-in-the-loop checkpoints; built a Fast Mode leveraging direct semantic search, letting users toggle between Fast Mode and Deep Mode.
- Designed business and technical architecture for the AI Portfolio POC/MVP end-to-end and demoed it to stakeholders and customers across multiple Airbus functions; leading productionization using React, FastAPI, pgvector/Cloud SQL, and Cloud Run with CI/CD, RBAC, and ReAct-based agent workflows.
- Joined the core AgentAIR team, working on building reusable platform capabilities, foundational components, and integration tools to streamline and accelerate the implementation of AI Agents across Airbus business units.
- Architected Airbus's innersourced central Skill Registry, a categorized library of agent, productivity, and platform skills (e.g. Claude Code skills) with references, reusable scripts, and skill creation templates; built a Jenkins CI/CD pipeline to publish and version skills to Artifactory, GCP Artifact Registry, and as Python packages.
- Reduced advanced search latency by 75% (2 min → 30 sec) through query optimization and efficient vector retrieval pipelines on Vertex AI; implemented prompt engineering best practices to minimize token usage and improve output accuracy.
- Cut Docker image sizes across Augmented BMS services, data ingestion by 50%, frontend by 80%, and backend by 10%, and optimized the chunking strategy (33K → 25K chunks), reducing Advanced Search query response time by 60%.
- Implemented hallucination detection and response grounding pipelines in Augmented BMS, identifying factually incorrect or unsupported outputs and anchoring responses to source documents, using RAGAS-based evaluation and benchmarking against golden datasets.
- Built usage insights and statistics dashboards for Augmented BMS (Google Data Studio) and AI Forge (React/Recharts), communicating weekly metrics and adoption statistics to stakeholders across Airbus functions.
- Built full GCP backend infrastructure using Cloud Run, Cloud Functions, Pub/Sub Triggers, Cloud SQL (PostgreSQL), Cloud Logging, and Terraform for IaC, enabling scalable and observable GenAI workloads.
- Developed the React.js frontend for Augmented BMS (conversational UI, advanced/general search, quick-action panels, thinking) and led frontend for AI Forge (Kaggle-like AI collaboration platform) with 50% UI ownership using React.js, Vite, Zustand, and Tailwind.
- Refined LLM inference quality by tuning generative configuration (top-k, top-p, temperature) and post-processing pipelines to structure responses and embed contextual document reference hyperlinks inline within chatbot outputs.
- Established testing strategy with Jest and Cypress within Jenkins CI/CD; mentored new team members on component architecture and GenAI integration best practices.

Airbus Group India Pvt Ltd. — **August 2022 – March 2025**
Associate Software Engineer — *Bengaluru, India*

- Owned and led Vue.js frontend for ModelOps Platform (official Airbus AI Platform), architecting Role Based Access Control and unifying Portal and Monitoring apps under a shell using Vite, Pinia, Chart.js, and a custom Design System.
- Designed architecture and led frontend for ML model creation, SageMaker deployment, and real-time monitoring with drift detection, alerts, and AWS cost/resource optimization dashboards; deployed infrastructure via CloudFormation and Python CDK.

- Co-developed Vue 3 Design System with 50% ownership, building 50+ configurable and reusable components with Storybook documentation aligned with Airbus UI/UX standards.
- Drove official adoption of Vue.js into Airbus engineering guidelines, expanding the component ecosystem beyond React and Angular, enabling faster development and standardization across multiple teams.
- Implemented Jest and Cypress testing within Jenkins CI/CD; bootstrapped frontend clones from ModelOps codebase to accelerate delivery for projects like GraphOps.

Achievements

SPOT Award (November 2025) — Airbus
Recognized for significant contributions to the Augmented BMS development.

SPOT Award (March 2025) — Airbus
Recognized for significant contributions to ModelOps Platform, the official AI Platform of Airbus.

Professional ML Engineer — Google Cloud
Certified in designing, building, and productionizing ML models to solve business challenges using Google Cloud technologies.

Advanced Generative AI: Deep Dive (2024) — Google Cloud
Hands-on training in building, tuning, and deploying generative AI models using Vertex AI.

Technical Skills

Languages & Frameworks: Python, FastAPI, SQL, JavaScript, TypeScript
GenAI, NLP & LLM: Google ADK, LangChain, Vertex AI, Multi-Agent Systems, ReAct Patterns, Human-in-the-Loop, Semantic Search, RAG, pgvector, Hallucination Detection, Benchmarking, RAGAS, LangSmith, Prompt Engineering, Pytorch, TensorFlow, LLMOps, MLOps vLLM, LangGraph, Voice AI, Model Context Protocol, Transformers, AI Evaluation.
Cloud — GCP: Cloud Run, Cloud Functions, Cloud Build, Pub/Sub Triggers, Cloud SQL (PostgreSQL), Cloud Logging, BigQuery, Looker Studio (Data Studio), Terraform, IAM
Cloud — AWS: Lambda, S3, DynamoDB, SageMaker, CDK, CloudFormation, VPC, IAM, ECS
Frontend: Vue.js, React.js, Vite, Zustand, Pinia, Tailwind CSS, Chart.js, Recharts, Storybook
Tools: Git, GitHub, JIRA, Jenkins, Docker, Cypress, Jest, Vitest, Webpack, Figma, Jupyter, Hugging face, Comfy UI